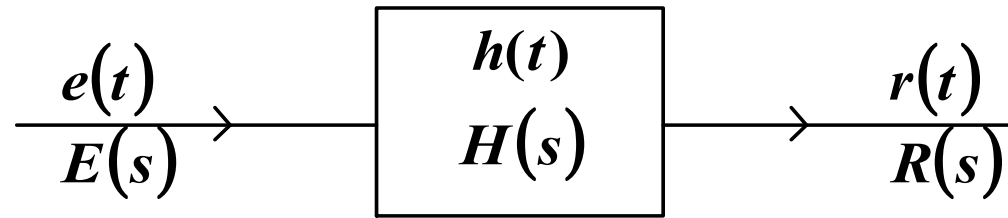


Mian Content

- **Definition of System Function**
- **Transformation of Continuous System Models**
- **Interconnection of Continuous Systems**

1. Definition of System Function



Time-domain analysis $r(t) = e(t) * h(t)$

Let $E(s) = \mathcal{L}[e(t)]$, $H(s) = \mathcal{L}[h(t)]$, then

$$\mathcal{L}[r(t)] = R(s) = E(s) \cdot H(s)$$

Definition: system function $H(s) = \frac{\text{Laplace transform of the zero-state response}}{\text{Laplace transform of the input signal}}$

$$H(s) = \frac{R(s)}{E(s)}$$

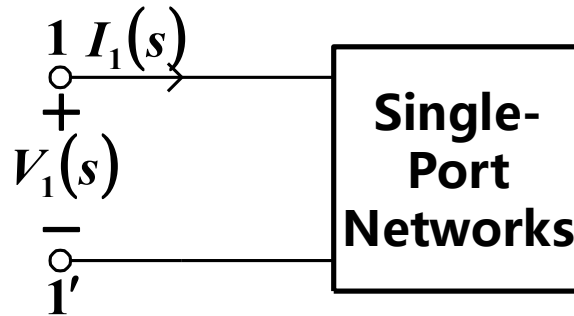
2. Transformation of Continuous System Models

The system function serves as a bridge for converting between different representations of continuous systems:

- **Circuit $\rightarrow$ s -Domain Equivalent Model**
- **Differential Equation $\rightarrow$ System Function $H(s)$**
- **Interconnected Systems**

(1) Determining System Functions from Circuit Diagrams

Driving-Point Functions: excitation and response are measured at the same port

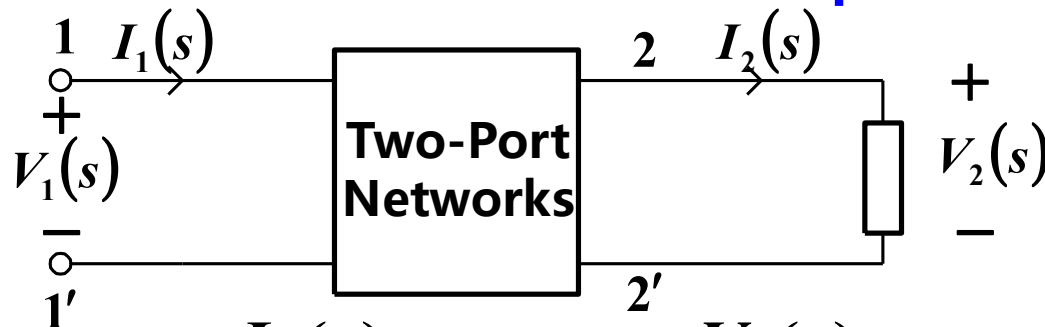


$$H(s) = \frac{I_1(s)}{V_1(s)}$$

$$H(s) = \frac{V_1(s)}{I_1(s)}$$

Driving-Point Admittance Driving-Point Impedance

Transfer Functions: excitation and response are measured at different ports



$$H(s) = \frac{I_2(s)}{V_1(s)}$$

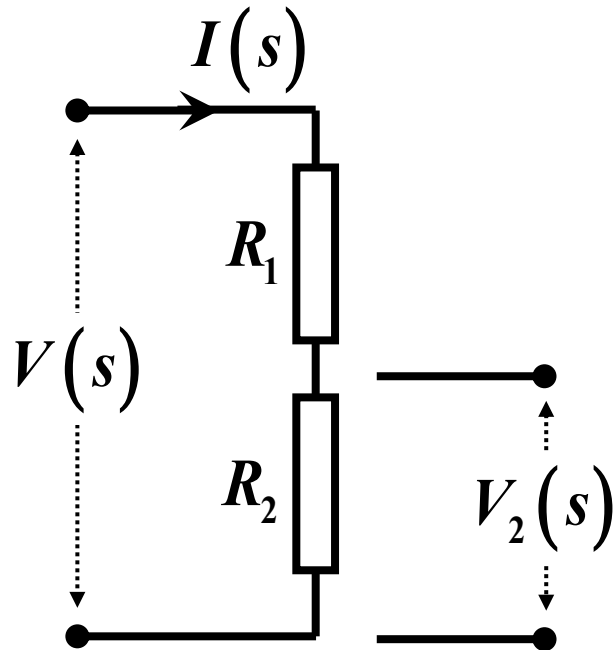
$$H(s) = \frac{V_2(s)}{I_1(s)}$$

$$H(s) = \frac{V_2(s)}{V_1(s)}$$

$$H(s) = \frac{I_2(s)}{I_1(s)}$$

Transfer Admittance Transfer Impedance Voltage Transfer Ratio Current Transfer Ratio

Example 1



(1) Driving-Point Admittance

Excitation: Voltage $V(s)$

Response: Current $I(s)$

The driving-point **admittance** is defined as:

$$H_1(s) = \frac{I(s)}{V(s)} = \frac{1}{R_1 + R_2}$$

(2) Voltage Transfer Function

Excitation: Voltage $V(s)$

Response: Voltage $V_2(s)$ across R_2

The **voltage transfer function** is:

$$H_2(s) = \frac{V_2(s)}{V(s)} = \frac{R_2}{R_1 + R_2}$$

(2) Deriving System Functions from Differential Equations

$$\sum_{k=0}^n a_k \frac{d^k r(t)}{dt^k} = \sum_{j=0}^m b_j \frac{d^j e(t)}{dt^j}$$

Taking the **bilateral** Laplace transform of both sides and applying the time-domain differentiation property of Laplace transforms yields

$$\sum_{k=0}^n a_k s^k R(s) = \sum_{j=0}^m b_j s^j E(s)$$

System Function Derivation

$$H(s) = \frac{R(s)}{E(s)} = \frac{\sum_{j=0}^m b_j s^j}{\sum_{k=0}^n a_k s^k}$$

Example 2: Solving for System Function and Unit Impulse Response of a Causal System

$$\frac{d^2 r(t)}{dt^2} + 5 \frac{dr(t)}{dt} + 6r(t) = \frac{d^2 e(t)}{dt^2} + \frac{de(t)}{dt}$$

Apply the **bilateral Laplace transform** to both sides of the equation (assuming zero initial conditions)

$$s^2 R(s) + 5sR(s) + 6R(s) = s^2 E(s) + sE(s)$$

$$(s^2 + 5s + 6)R(s) = (s^2 + s)E(s)$$

$$H(s) = \frac{R(s)}{E(s)} = \frac{s^2 + s}{s^2 + 5s + 6}$$

Example 2

$$H(s) = \frac{s^2 + s}{s^2 + 5s + 6} = 1 - \frac{4s + 6}{s^2 + 5s + 6}$$

$$\text{let } \frac{4s + 6}{s^2 + 5s + 6} = \frac{k_1}{s + 2} + \frac{k_2}{s + 3}$$

$$k_1 = (s + 2) \frac{4s + 6}{s^2 + 5s + 6} \Big|_{s=-2} = -2$$

$$k_2 = (s + 3) \frac{4s + 6}{s^2 + 5s + 6} \Big|_{s=-3} = 6$$

$$H(s) = 1 + \frac{2}{s + 2} - \frac{6}{s + 3}$$

$$\text{Thus } h(t) = \delta(t) + 2e^{-2t} u(t) - 6e^{-3t} u(t)$$

(3) Zeros and Poles of System Functions

Factored form

$$H(s) = \frac{B(s)}{A(s)} = K \frac{(s - z_1)(s - z_2) \cdots (s - z_j) \cdots (s - z_m)}{(s - p_1)(s - p_2) \cdots (s - p_k) \cdots (s - p_n)}$$

$$= K \frac{\prod_{j=1}^m (s - z_j)}{\prod_{k=1}^n (s - p_k)}$$

Zeros $z_1, z_2 \cdots z_n$ make $H(s)=0$

Poles $p_1, p_2 \cdots p_n$ make $H(s)$ unbounded

K: Scalar multiplier

Zero-Pole Plot in the s-Plane:

Poles: Represented by $\times$ (cross)

Zeros: Represented by $\bigcirc$ (circle)

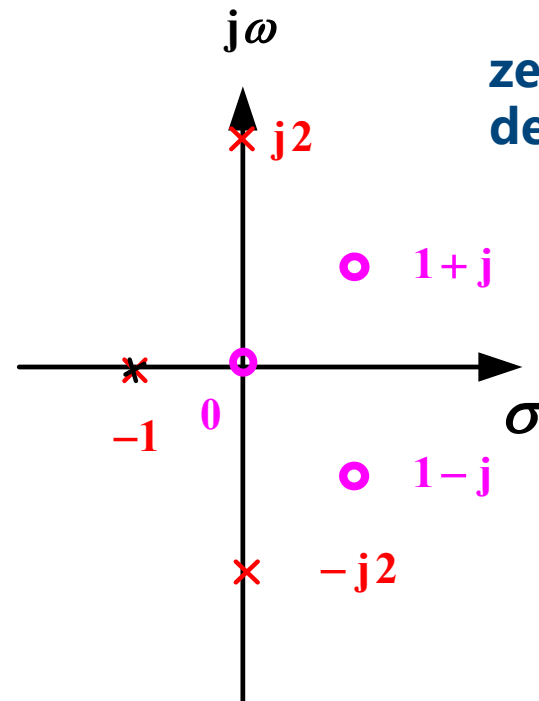
Example 3

$$H(s) = \frac{s(s-1+j1)(s-1-j1)}{(s+1)^2(s+j2)(s-j2)}$$

Poles: $p_1 = p_2 = -1$, $p_3 = -j2$, $p_4 = j2$

Zeros: $z_1 = 0$, $z_2 = 1-j1$, $z_3 = 1+j1$, $z_4 = \infty$

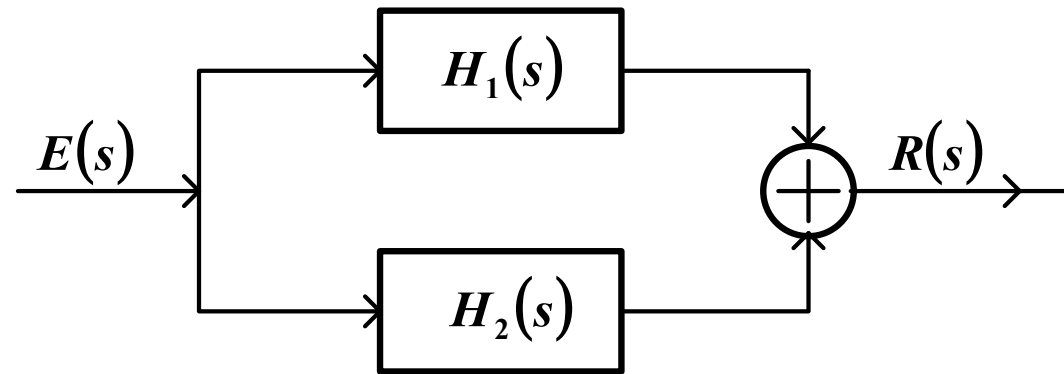
Zero-Pole Plot:



zero at infinity, due to numerator degree < denominator

3. Interconnected Systems

(1) Parallel Connection of LTI Systems

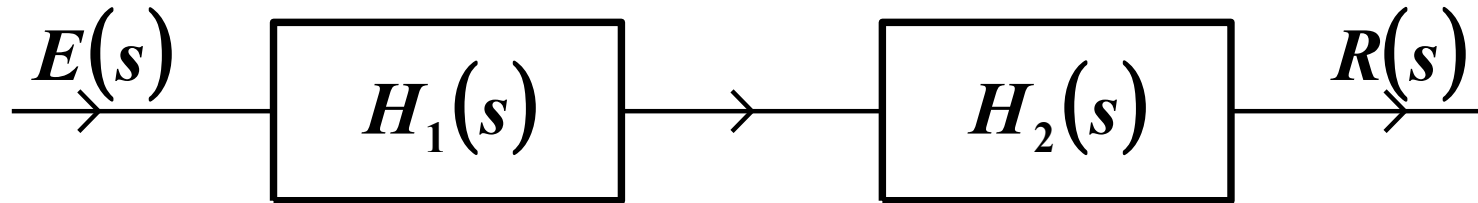


$$\begin{aligned} R(s) &= E(s) H_1(s) + E(s) H_2(s) \\ &= E(s) H(s) \end{aligned}$$

$$H(s) = H_1(s) + H_2(s)$$

3. Interconnected Systems

(2) Cascade Connection of LTI Systems

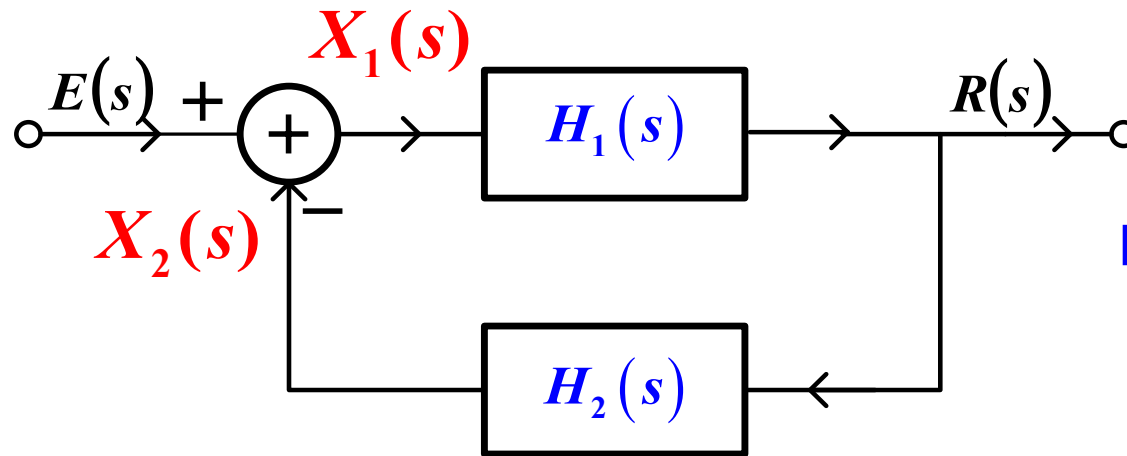


$$\begin{aligned} R(s) &= E(s) H_1(s) \cdot H_2(s) \\ &= E(s) \cdot H(s) \end{aligned}$$

$$H(s) = H_1(s) \cdot H_2(s)$$

3. Interconnected Systems

(3) LTI Feedback Systems



Adder Equation

$$X_1(s) = E(s) - X_2(s)$$

Forward Path

$$R(s) = X_1(s)H_1(s)$$

Feedback Path

$$X_2(s) = R(s) \cdot H_2(s)$$

3. Interconnected Systems

(3) LTI Feedback Systems

$$X_1(s) = E(s) - X_2(s)$$

$$R(s) = X_1(s)H_1(s)$$

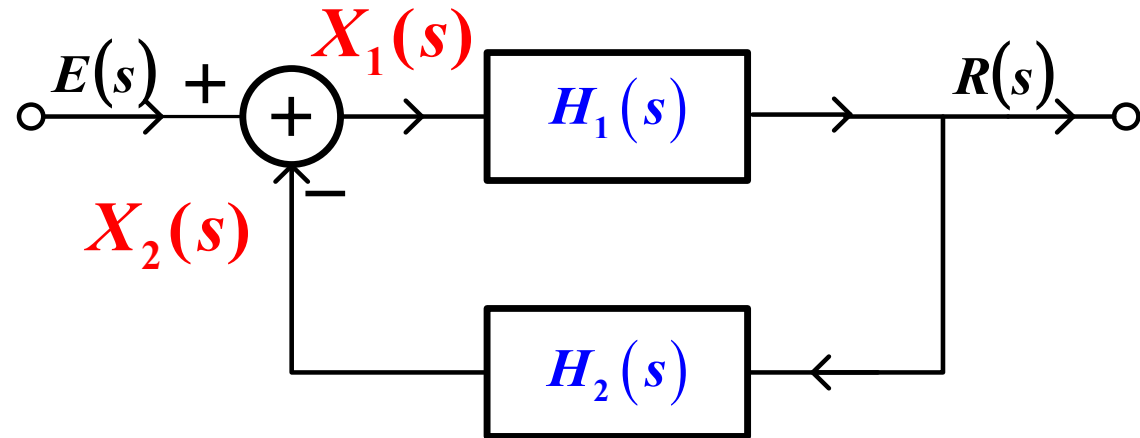
$$X_2(s) = R(s) \cdot H_2(s)$$

Substitution

$$\frac{R(s)}{H_1(s)} = E(s) - R(s) \cdot H_2(s)$$

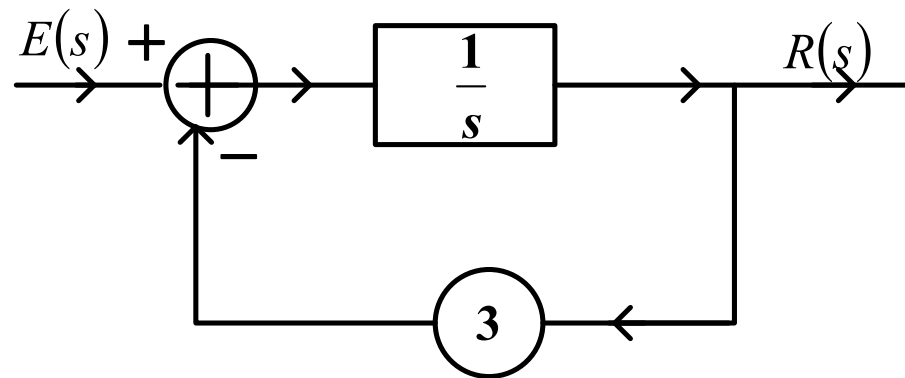
Closed-loop transfer function

$$H(s) = \frac{R(s)}{E(s)} = \frac{H_1(s)}{1 + H_1(s)H_2(s)}$$



Example 4

Given the system block diagram below, derive the system function and the differential equation describing this system



$$H_1(s) = \frac{1}{s} \quad H_2(s) = 3$$

$$H(s) = \frac{R(s)}{E(s)} = \frac{H_1(s)}{1 + H_1(s)H_2(s)} = \frac{1}{s + 3}$$

Example 4

$$H(s) = \frac{R(s)}{E(s)} = \frac{1}{s+3}$$

s-Domain Equation: the relationship between excitation and response

$$sR(s) + 3R(s) = E(s)$$

Time-Domain Differential Equation: Apply the inverse Laplace transform

$$\frac{dr(t)}{dt} + 3r(t) = e(t)$$

Summary

- Definition of System Function
- Transformation of Continuous System Models
- Interconnection of Continuous Systems

$$H(s) = \frac{R(s)}{E(s)}$$

Main Content

- **Assessing System Stability via Impulse Response**
- **Determining System Stability from Zero-Pole Distribution of the System Function**

1. Problem Statement

Given the transfer function of a continuous-time system:

$$H(s) = \frac{1}{s+1} + \frac{0.001}{s-2}$$

When the input is a unit step signal $u(t)$, the Laplace transform of the zero-state response is:

$$\begin{aligned} R_{zs}(s) &= \frac{H(s)}{s} \\ &= \frac{1 - 0.0005}{s} - \frac{1}{s+1} + \frac{0.0005}{s-2} \end{aligned}$$

$$r_{zs}(t) = \left(1 - e^{-t} + 0.0005 e^{2t}\right) u(t)$$

Though $0.0005 \ll 1...$

System Instability: The presence of a right half-plane (RHP) pole at $s=2$ makes the system unstable

Engineering Insight: Even with negligible coefficients for unstable modes, long-term behavior is governed by the fastest-growing exponential term

2. General Criteria for System Stability

(1) Definition:

A system is said to be **Bounded-Input Bounded-Output (BIBO) stable** (or simply stable) if, for any **bounded input**, its zero-state **response** is also **bounded**

Necessary and Sufficient Condition for Stability (Absolute Integrability Condition):

$$\int_{-\infty}^{\infty} |h(t)| dt \leq M$$

where M is a finite positive constant

2. General Criteria for System Stability

Time-Domain Stability Criterion: $\int_{-\infty}^{\infty} |h(t)| dt \leq M$

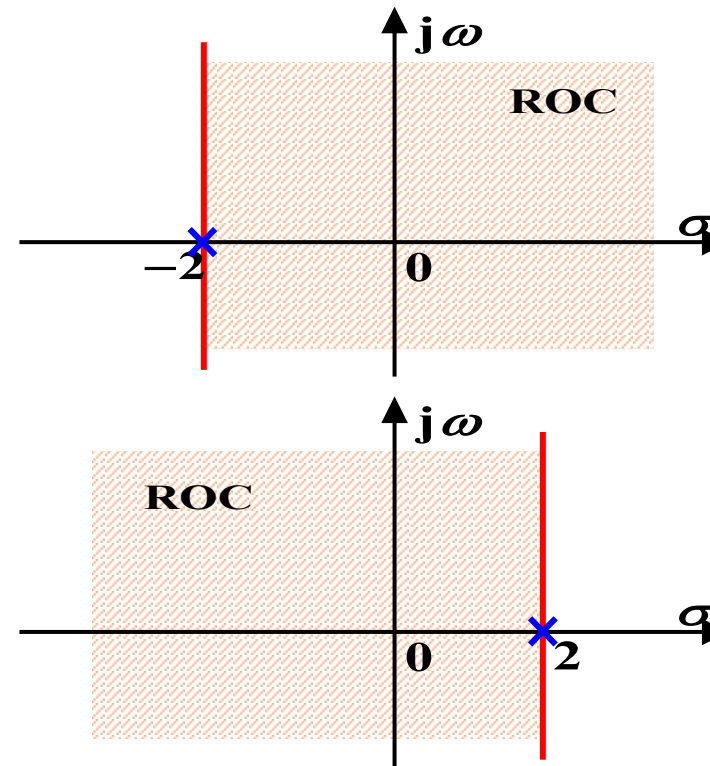
s-Domain Stability Criterion: A system is stable if the **ROC** of its transfer function $H(s) = \mathcal{L}[h(t)]$ **includes the imaginary axis**

$$h(t) = e^{-2t} u(t)$$

$$H(s) = \frac{1}{s+2} \quad (\sigma > -2)$$

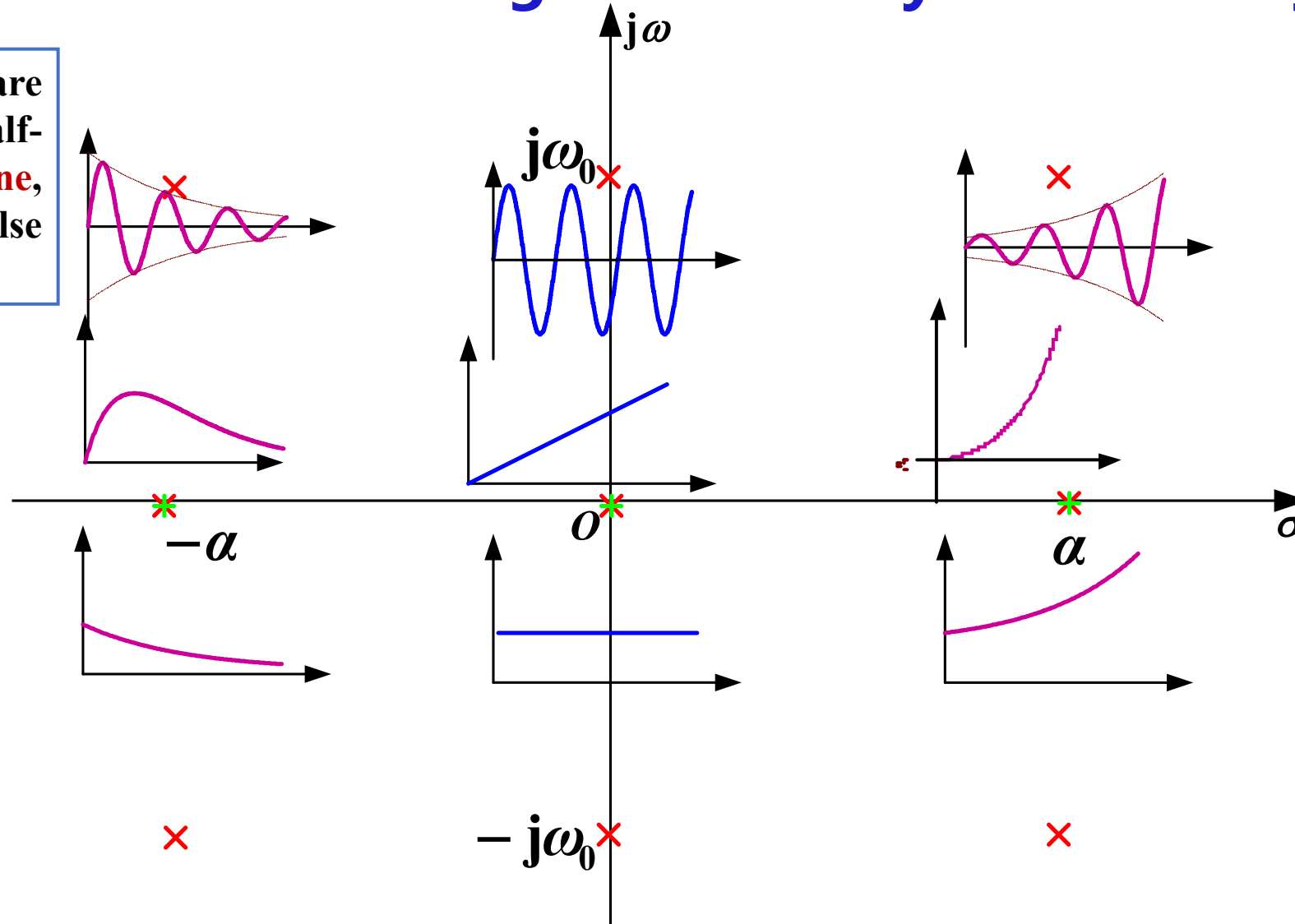
$$h(t) = e^{2t} u(-t)$$

$$H(s) = -\frac{1}{s-2} \quad (\sigma < 2)$$



3. Method for Determining the Stability of Causal Systems

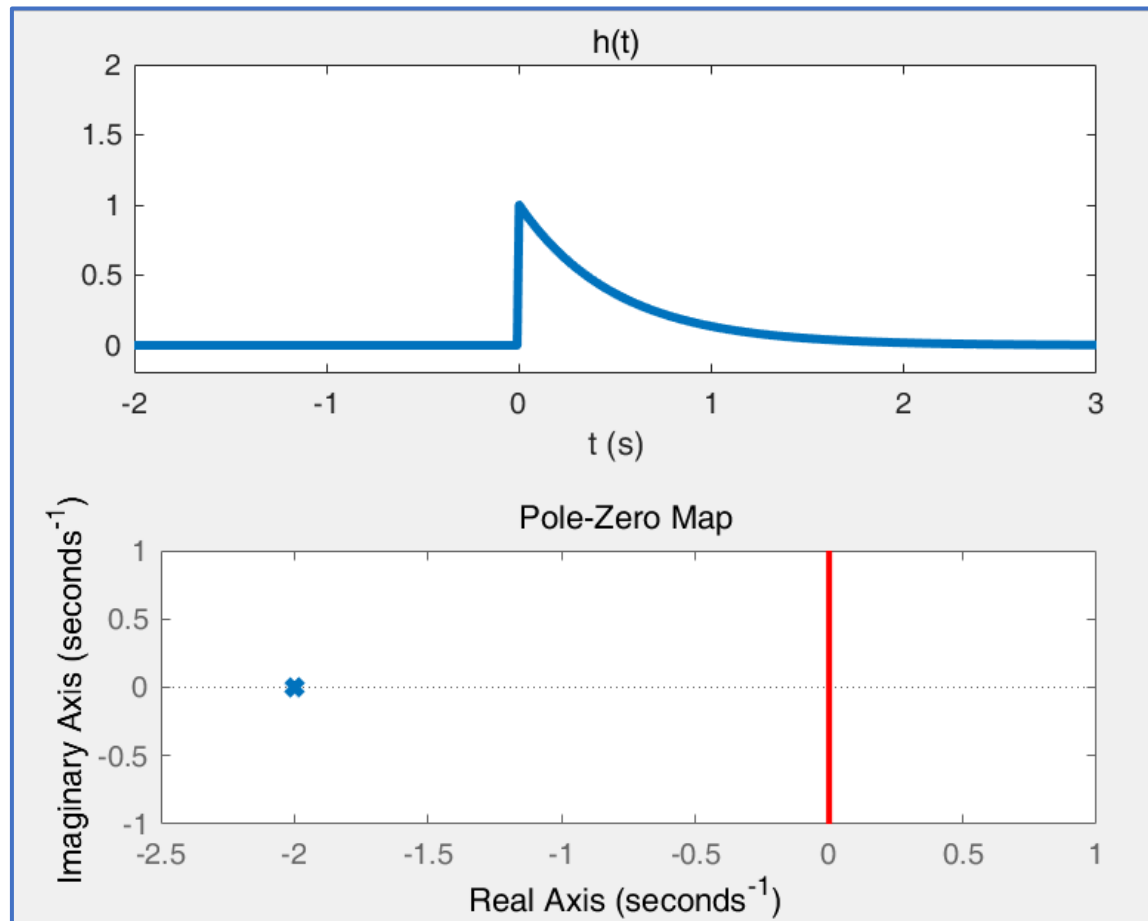
When all poles are located in the **left** half-plane of the **s-plane**, the unit impulse response **converges**



3. Method for Determining the Stability of Causal Systems

Single Real Pole

$$H(s) = \frac{1}{s + \alpha} \quad h(t) = e^{-\alpha t} u(t)$$

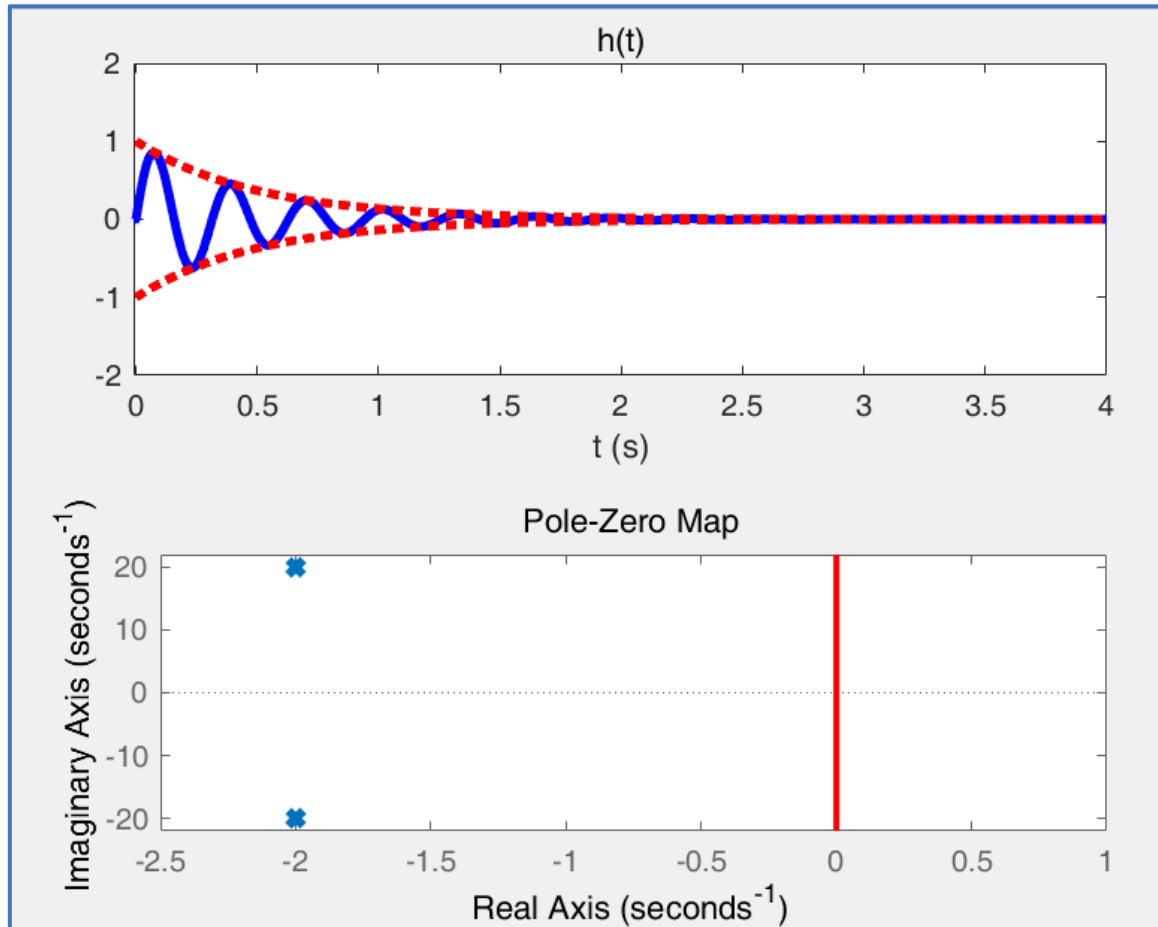


- Pole on the **Left** Real Axis
 $h(t)$ Exponential **decay**
- Pole at the **Origin**
 $h(t)$ **Constant**
- Pole on the **Right** Real Axis
 $h(t)$ Exponential **growth**

3. Method for Determining the Stability of Causal Systems

Conjugated complex poles

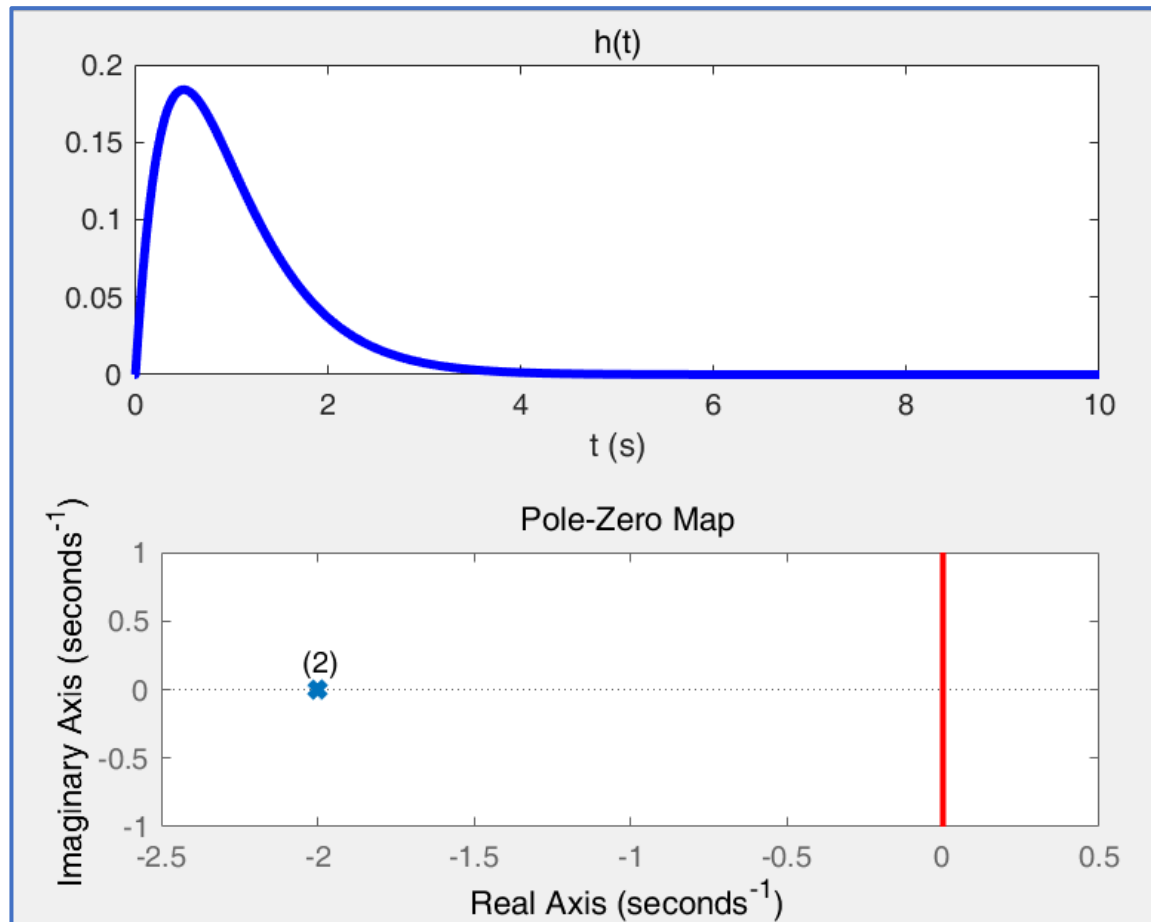
$$H(s) = \frac{\omega_0}{(s + \alpha)^2 + \omega_0^2} \quad h(t) = e^{-\alpha t} \sin(\omega_0 t) \cdot u(t)$$



- Pole in the **Left** Half-Plane
 $h(t)$ **Decaying** oscillation
- Pole on the **Imaginary** Axis
 $h(t)$ **Sustained** oscillation
- Pole in the **Right** Half-Plane
 $h(t)$ **Growing** oscillation

3. Method for Determining the Stability of Causal Systems

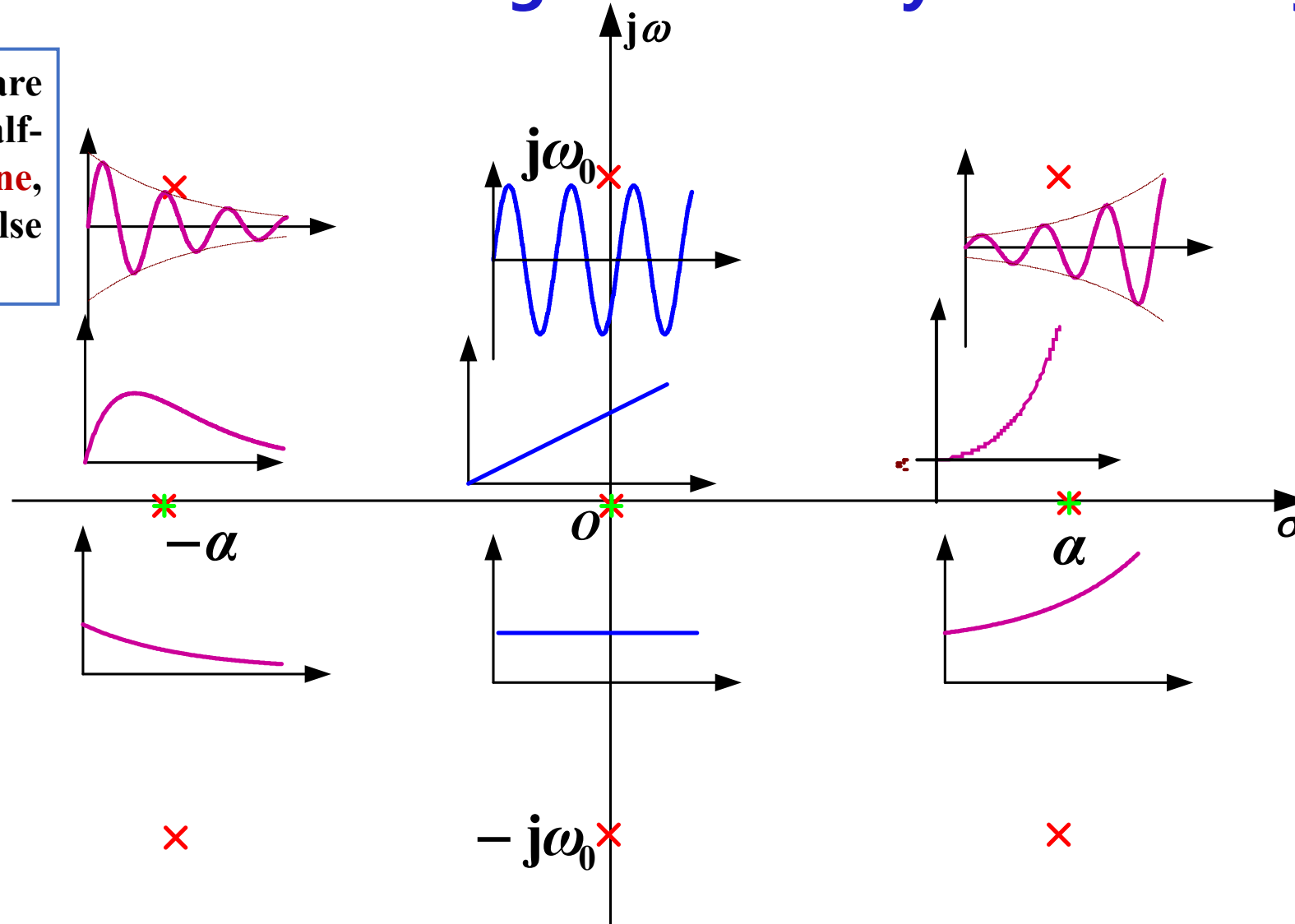
Repeated poles $H(s) = \frac{1}{(s + \alpha)^2}$ $h(t) = t e^{-\alpha t} u(t)$



- Pole on the **Left** Real Axis
 $h(t)$ **decaying** oscillation
- Pole at the **Origin**
 $h(t)$ **linear growth** (marginally unstable)
- Pole on the **Right** Real Axis
 $h(t)$ **Exponential growth**

3. Method for Determining the Stability of Causal Systems

When all poles are located in the **left** half-plane of the **s-plane**, the unit impulse response **converges**



3. Method for Determining the Stability of Causal Systems

(1) Stable System

A system is stable if all poles of $H(s)$ lie strictly in the **left half-plane** (LHP) of the s -plane (*excluding* the imaginary axis). This ensures:

$$\lim_{t \rightarrow \infty} h(t) = 0 \text{ (Impulse response decays to zero)}$$

(2) Unstable System

A system is unstable if: Any pole lies in **the right half-plane** (RHP), or there are repeated poles (second-order or higher) on the imaginary axis,

$$\lim_{t \rightarrow \infty} h(t) \rightarrow \infty \text{ (Impulse response diverges)}$$

(3) Marginally Stable (Technically Unstable) System

A system is marginally stable if: Poles are simple (first-order) and lie exactly on the imaginary axis

$$\text{As } t \rightarrow \infty, h(t) \text{ approaches } a \begin{cases} \text{non-zero constant,} \\ \text{or sustained oscillation.} \end{cases}$$

7.2 Stability of Continuous-Time Systems

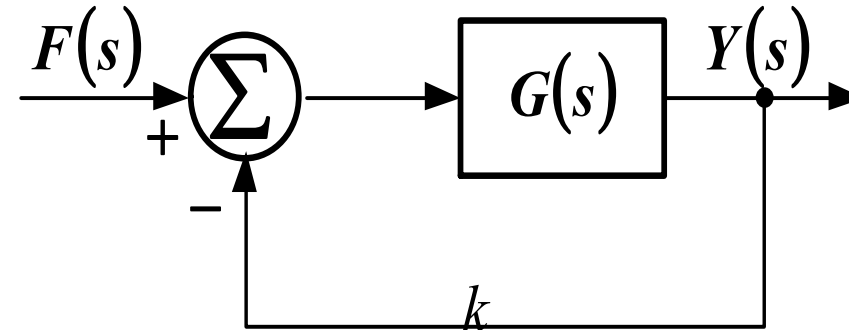
Example 1

A causal feedback system is shown in the figure

The transfer function of the subsystem is

$$G(s) = \frac{1}{(s-1)(s+2)}$$

This subsystem is unstable



Under what condition of the constant k is the system stable?

Based on the given conditions, the equation is derived as:

$$H(s) = \frac{Y(s)}{F(s)} = \frac{G(s)}{1 + kG(s)} = \frac{1}{s^2 + s - 2 + k}$$

7.2 Stability of Continuous-Time Systems

Example 1

$$H(s) = \frac{1}{s^2 + s - 2 + k}$$

The poles of $H(s)$

$$p_{1,2} = -\frac{1}{2} \pm \sqrt{\frac{9}{4} - k}$$

For all poles to lie in the left half-plane of s , the following must hold

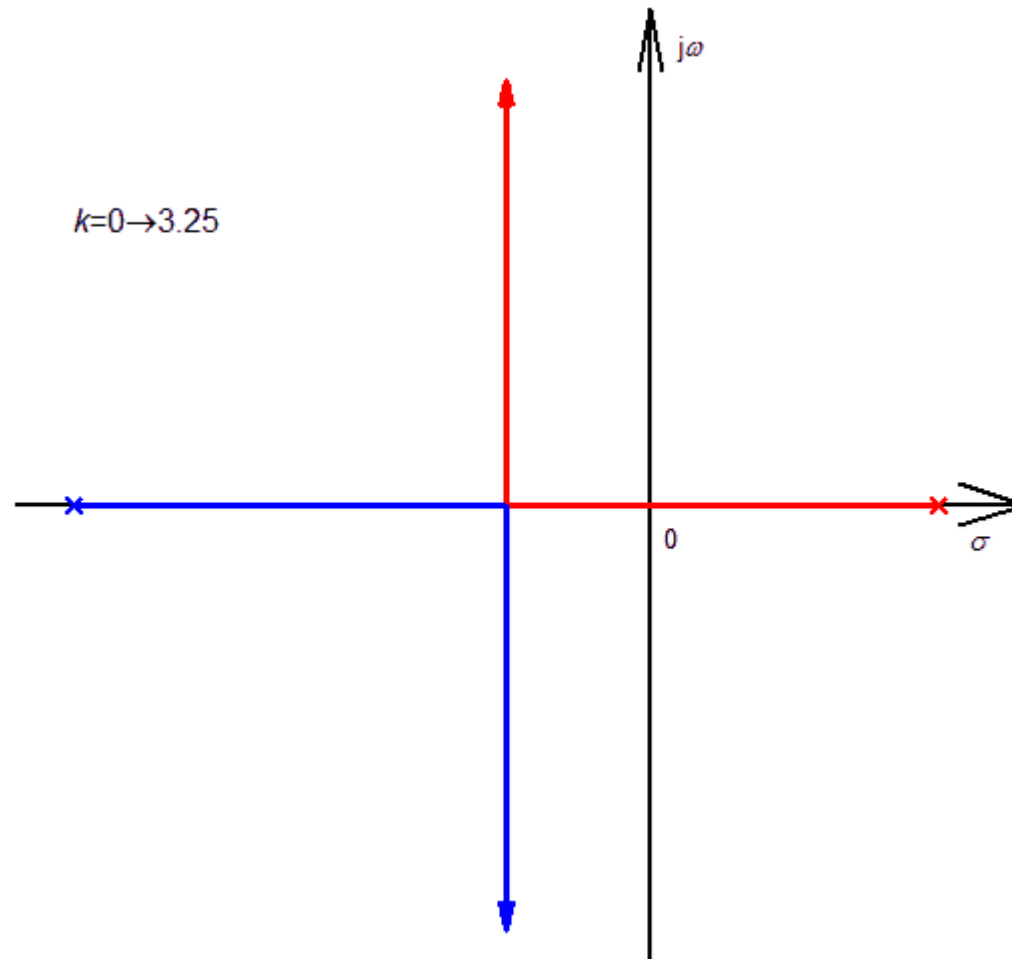
$$\frac{9}{4} - k < 0 \quad \text{or} \quad \begin{cases} \frac{9}{4} - k \geq 0 \\ -\frac{1}{2} + \sqrt{\frac{9}{4} - k} < 0 \end{cases}$$

Solving these conditions yields $k > 2$. Therefore, the system is stable when $k > 2$

7.2 Stability of Continuous-Time Systems

Example 1

By introducing **negative feedback**, the **unstable system achieves stability**



Summary

BIBO Stability:

A system is called **Bounded-Input Bounded-Output (BIBO) stable**, or simply **stable**, if for any bounded input, its zero-state response is also bounded

Time-Domain Criterion:

$$\int_{-\infty}^{\infty} |h(t)| dt \leq M$$

(where M is a finite constant)

s-Domain Criterion: For $H(s) = L[h(t)]$, the system is stable if the region of convergence **includes the imaginary axis**

Causal and stable: **All poles of $H(s)$ lie in the left half of the s-plane**

1. Introduction

When introducing the Laplace transform, we addressed cases where $f(t)$ fails to satisfy the absolute integrability condition. By multiplying it with a decaying factor $e^{-\sigma t}$ and applying the Fourier transform, we derived the Laplace transform:

$$\mathcal{F}[f(t) \cdot e^{-\sigma t} u(t)] = \int_{0_-}^{\infty} [f(t) \cdot e^{-\sigma t}] e^{-j\omega t} dt$$

$$s = \sigma + j\omega$$

$$\mathcal{L}[f(t)] = F(s)$$

2. Relationship Between FT and LT

(1) When $\sigma_0 > 0$, the convergence boundary lies in the **right** half of the s -plane

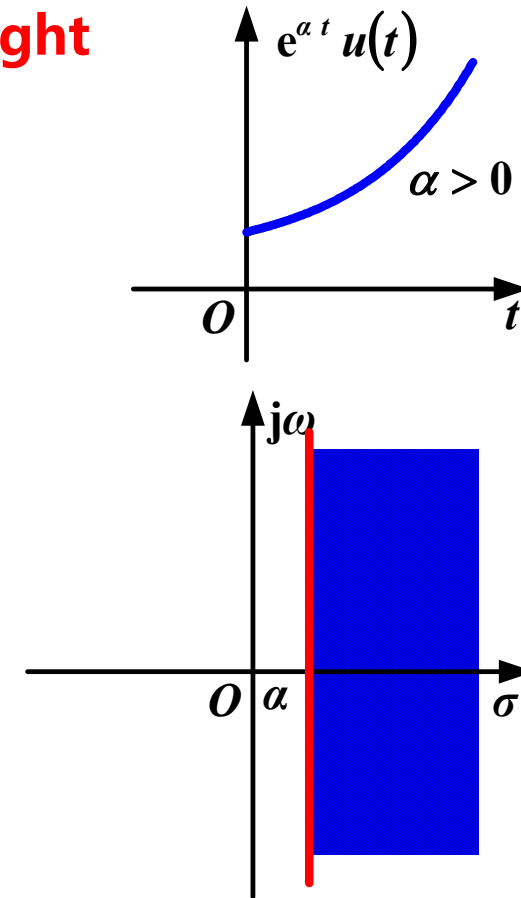
Consider the function:

$$f(t) = e^{\alpha t} u(t) (\alpha > 0)$$

Its **Laplace transform** is:

$$F(s) = \frac{1}{s - \alpha}$$

- Region of Convergence (**ROC**): $\sigma > \alpha$
- Fourier Transform Note:
 $F(j\omega)$ does **not exist** in this case
Thus, $F(j\omega)$ cannot be derived from $F(s)$



2. Relationship Between FT and LT

(2) When $\sigma_0 < 0$, the convergence boundary lies in the **left** half of the s -plane

Consider the decaying exponential function:

$$f(t) = e^{-\alpha t} u(t) (\alpha > 0)$$

Its Laplace transform is:

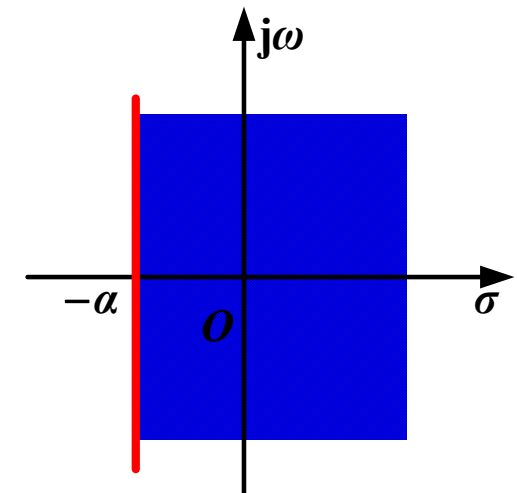
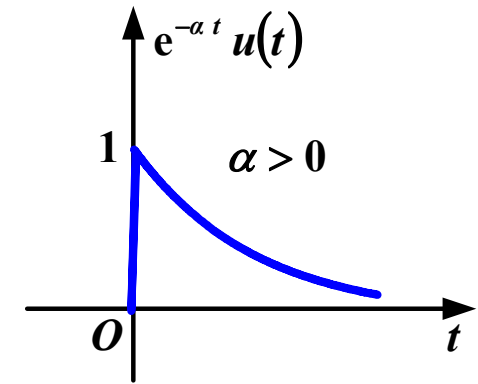
$$F(s) = \frac{1}{\alpha + s}$$

- Region of Convergence (ROC): $\sigma > -\alpha$

- Fourier Transform Existence:

Since $f(t)$ is a decaying function, its Fourier transform exists and can be obtained by evaluating $F(s)$ at $s = j\omega$:

$$F(j\omega) = F(s) |_{s=j\omega} = \frac{1}{\alpha + j\omega}$$

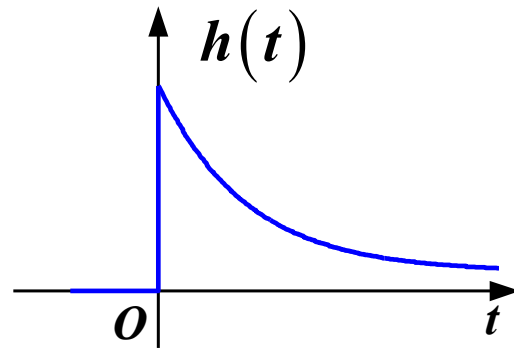


8 Relationship Between LT and FT

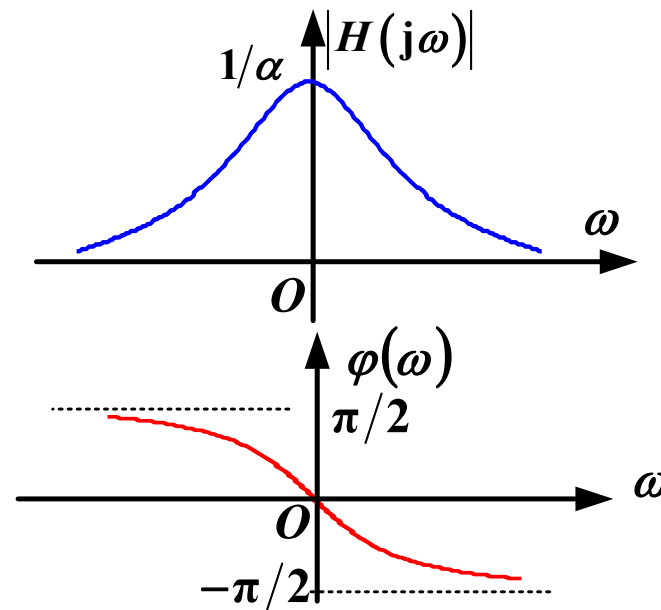
2. Relationship Between FT and LT

$$h(t) = e^{-\alpha t} \quad (\alpha > 0) \quad H(j\omega) = \frac{1}{j\omega + \alpha}$$

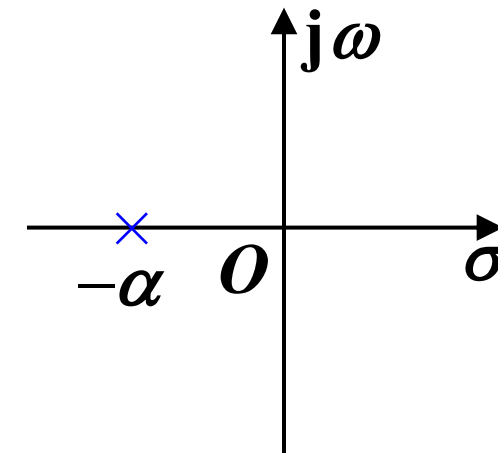
$$H(s) = \frac{1}{s + \alpha}$$



Time Domain



Frequency Domain



Pole-Zero Plot

2. Relationship Between FT and LT

(3) When $\sigma_0 = 0$, the convergence boundary lies on the **imaginary axis**

Key Observations: $F(s)$ exists, but $F(j\omega)$ cannot be obtained by simply substituting $s = j\omega$ in $F(s)$, as the Fourier transform includes singular function terms

Example

$$f(t) = u(t)$$

$$F(s) = \frac{1}{s}$$

$$F(j\omega) = \pi \delta(\omega) + \frac{1}{j\omega}$$

Summary

Relationship Between $F(s)$ and $F(j\omega)$:

Case 1: $\sigma_0 > 0$

Convergence axis lies in the right half of the s -plane, $F(j\omega)$ does not exist

Case 2: $\sigma_0 < 0$

Convergence axis lies in the left half of the s -plane, direct substitution is valid

$$F(j\omega) = F(s) |_{s=j\omega}$$

Case 3: $\sigma_0 = 0$

Convergence axis lies on the imaginary axis,

$$F(j\omega) = F(s) |_{s=j\omega} + \pi \sum_n k_n \delta(\omega - \omega_n)$$

Requires inclusion of impulse terms (Dirac delta functions)



香港科技大學
THE HONG KONG UNIVERSITY OF SCIENCE AND TECHNOLOGY

Thank you for listening !

